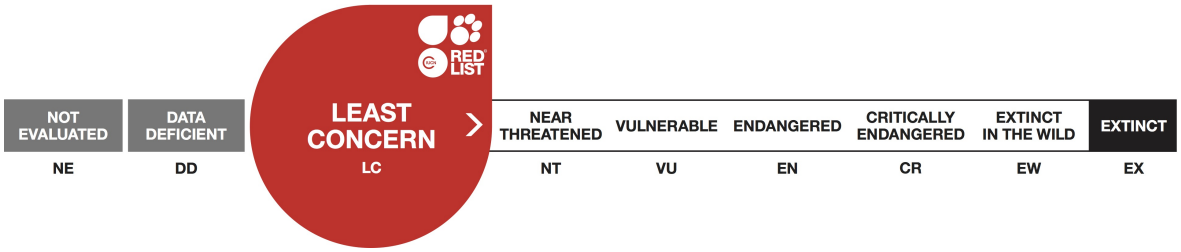
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Sepia officinalis, Common Cuttlefish

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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Mollusca	Cephalopoda	Sepioloida	Sepiidae

Taxon Name: *Sepia officinalis* Linnaeus, 1758

Synonym(s):

- *Sepia filliouxi*
- *Sepia mediterranea*

Common Name(s):

- English: Common Cuttlefish
- French: Casseron, Chakod, Chibia, Chubei, Margade, Seiche, Seiche commune, Supia
- Spanish: Aluda, Choco, Coca, Jibia, Jibión, Luda, Rellena, Relleno, Sepia común, Sipia, Sipionet

Assessment Information

Red List Category & Criteria: Least Concern ver 3.1

Year Published: 2012

Date Assessed: March 15, 2009

Justification:

Sepia officinalis has been assessed as Least Concern as although it is the focus of large-scale commercial fisheries, particularly in the Mediterranean Sea and off the west coast of Africa and may be close to being overexploited in some regions (e.g. the Mediterranean Sea), it has a wide geographic range and is thus likely to not be threatened. Furthermore, FAO statistics indicate a constant yield of approximately 15,000 tonnes per annum suggesting no overall decline in stocks. We therefore consider this species to be of Least Concern.

Geographic Range

Range Description:

This species has a wide geographic distribution (Reid *et al.* 2005). It occurs in the northeast and east Atlantic Ocean and Mediterranean Sea extending from the Shetland Islands and Norway in the north, through the Mediterranean Sea to northwest Africa (i.e. to Senegal) in the south (Reid *et al.* 2005). It is not present in the Baltic Sea (Reid *et al.* 2005).

Country Occurrence:

Native: Albania; Algeria; Belgium; Bosnia and Herzegovina; Croatia; Cyprus; Denmark; Egypt (Egypt (African part), Sinai); France (Corsica, France (mainland)); Germany; Greece (East Aegean Is., Greece (mainland), Kriti); Ireland; Israel; Italy (Italy (mainland), Sardegna, Sicilia); Lebanon; Libya; Mauritania; Montenegro; Morocco; Netherlands; Norway; Portugal (Madeira, Portugal (mainland)); Senegal; Slovenia; Spain (Baleares, Canary Is., Spain (mainland), Spanish North African Territories); Sweden; Tunisia; Turkey (Turkey-in-Asia, Turkey-in-Europe); United Kingdom (Great Britain, Northern Ireland); Western Sahara

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FAO Marine Fishing Areas:

Native: Atlantic - eastern central, Atlantic - northeast, Mediterranean and Black Sea -

Population

Although estimates of the total population size do not exist, stock assessments of the exploited English Channel stock have been made (Royer *et al.* 2006). In spite of some growth overfishing, there was no indication that the species was at risk. Some assumptions of this paper were not met, so the results should be considered with care, but since 2006 cuttlefish landings from the English Channel have remained high suggesting they are being exploited at sustainable levels.

Current Population Trend: Unknown

Habitat and Ecology (see Appendix for additional information)

The common cuttlefish is a large species and can attain a maximum mantle length of 490 mm and body weight of 2 kg in temperate waters, and 300 mm and 2 kg in subtropical regions (Reid *et al.* 2005). It inhabits sandy or muddy substrates and can tolerate brackish waters; younger individuals tolerate lower salinities and more environmental instability than adults (Reid *et al.* 2005). Both adults and young bury in the sand during the day (Reid *et al.* 2005). It ambushes prey from its hiding place in the sand, feeding on a wide variety of prey including crustaceans, molluscs, polychaetes, small demersal fish as well as other cuttlefish (cannibalism is common when other prey abundances are low) (Reid *et al.* 2005). They are preyed upon by sharks, demersal fishes and other cephalopods (Reid *et al.* 2005). Growth rates are rapid leading to a life span of one to two years (Reid *et al.* 2005). During autumn and winter individuals migrate to deeper water (approximately 100m); returning to shallow water in spring and summer (Reid *et al.* 2005). In the Mediterranean large males return to shallow waters ahead of females with females and smaller individuals joining them throughout the spring and summer (Reid *et al.* 2005). Males demonstrate courtship behaviour and will guard females from rival males (Reid *et al.* 2005). Spawning occurs in shallow, inshore waters in April to July in the western Mediterranean and January to April off Senegal (Reid *et al.* 2005). Males have on average 1,400 spermatophores, and females can have between 150 and 4,000 eggs (8 to 10mm in diameter) depending on their body size (Reid *et al.* 2005). The eggs are attached to a range of substrates, including seaweed and shells, and are darkened with ink (Reid *et al.* 2005). The duration of embryonic development is temperature dependent and ranges from 30 to 90 days (Reid *et al.* 2005). Those young that hatch in spring usually spawn in the autumn of the following year; those that hatch in autumn usually spawn in the spring of their second year (Reid *et al.* 2005). Young are restricted to shallow water until their cuttlebones are fully formed (Reid *et al.* 2005). Due to post spawning mortality in females there is sometimes a bias in adult males (Reid *et al.* 2005). This species has been raised successfully in aquaculture (Reid *et al.* 2005).

Systems: Marine

Use and Trade (see Appendix for additional information)

This species is fished intensively in the Mediterranean Sea and is a commercially important species in many countries (e.g. Greece, Spain and Tunisia) (Reid *et al.* 2005). Indeed, it may be near its sustainable limit in the Mediterranean (Reid *et al.* 2005). Highest catches are recorded for Tunisia in the Mediterranean, and for Spain and Morocco off west Africa (Reid *et al.* 2005). In the English Channel it has become increasingly targeted by the United Kingdom and France (Reid *et al.* 2005). It is caught using a variety of different gear types and females may be used as lures to trap males in the spawning season (Reid *et al.* 2005). It is also caught as bycatch (Reid *et al.* 2005). This species has potential for being raised in aquaculture, which may help prevent its overexploitation (Reid *et al.* 2005). It is exported to

Japan and Korea (Reid *et al.* 2005).

Threats (see Appendix for additional information)

Ocean acidification caused by increased levels of carbon dioxide in the atmosphere is potentially a threat to all cuttlefish. Studies have shown that under high pCO_2 concentrations, cuttlefishes actually lay down a denser cuttlebone which is likely to negatively affect buoyancy regulation (Gutowska *et al*. 2010). This species is a commercially important fisheries species in the Mediterranean Sea and off the west coast of Africa (Reid *et al.* 2005). It is intensively fished in the Mediterranean Sea and may be close to its sustainable limit (Reid *et al.* 2005). Females may also be used as lures to traps during the spawning season (Reid *et al.* 2005). It is also caught as bycatch (Reid *et al.* 2005).

Conservation Actions

No conservation measures are currently needed for this species and none are in place. Further research is recommended regarding the population trends, distribution, life history traits and threats impacting this species.

Credits

Assessor(s): Barratt, I. & Allcock, L.

Reviewer(s): Reid, A., Rogers, Alex & Bohm, M.

Contributor(s): Robin, J.-P., Herdson, R. & Duncan, C.

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External Resources

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Appendix

Habitats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Habitat	Season	Suitability	Major Importance?
9. Marine Neritic -> 9.3. Marine Neritic - Subtidal Loose Rock/pebble/gravel	-	Suitable	-
9. Marine Neritic -> 9.4. Marine Neritic - Subtidal Sandy	-	Suitable	-
9. Marine Neritic -> 9.5. Marine Neritic - Subtidal Sandy-Mud	-	Suitable	-
9. Marine Neritic -> 9.6. Marine Neritic - Subtidal Muddy	-	Suitable	-

Use and Trade

(http://www.iucnredlist.org/technical-documents/classification-schemes)

End Use	Local	National	International
Food - human	Yes	Yes	Yes

Threats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Threat	Timing	Scope	Severity	Impact Score
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.1. Intentional use: (subsistence/small scale)	Ongoing	Unknown	Unknown	Unknown
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.2. Intentional use: (large scale)	Ongoing	Unknown	Unknown	Unknown
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.3. Unintentional effects: (subsistence/small scale)	Ongoing	Unknown	Unknown	Unknown
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.4. Unintentional effects: (large scale)	Ongoing	Unknown	Unknown	Unknown
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration	Future	Unknown	Unknown	Unknown
11. Climate change & severe weather -> 11.5. Other impacts	Future	Unknown	Unknown	Unknown

Research Needed

(http://www.iucnredlist.org/technical-documents/classification-schemes)

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Research Needed
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
1. Research -> 1.5. Threats

Additional Data Fields

Distribution
Lower depth limit (m): 200
Population
Population severely fragmented: No
Habitats and Ecology
Movement patterns: Full Migrant

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